

SOLIDWORKS API (.NET VBA C#) — Reference (272/337): ISketchManager (part 2 of 3)

SOLIDWORKS API Documentation (converted from CHM)

ISketchManager (part 2 of 3) (Type)

SOLIDWORKS API Help

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CreateSketchBelt Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateSketchBelt Method (ISketchManager)

Creates a sketch belt feature.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSketchBelt( _  
    ByVal Length As System.Double, _  
    ByVal Thickness As System.Double, _  
    ByVal Engage As System.Integer, _  
    ByVal PulleyCount As System.Integer, _  
    ByVal Side1 As System.Integer, _  
    ByVal Side2 As System.Integer, _  
    ByVal Side3 As System.Integer, _  
    ByVal Side4 As System.Integer, _  
    ByVal Side5 As System.Integer, _  
    ByVal Side6 As System.Integer, _  
    ByVal Side7 As System.Integer, _  
    ByVal Side8 As System.Integer, _  
    ByVal Side9 As System.Integer, _  
    ByVal Side10 As System.Integer, _  
    ByVal Side11 As System.Integer, _  
    ByVal Side12 As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim Length As System.Double  
Dim Thickness As System.Double  
Dim Engage As System.Integer  
Dim PulleyCount As System.Integer  
Dim Side1 As System.Integer  
Dim Side2 As System.Integer  
Dim Side3 As System.Integer  
Dim Side4 As System.Integer  
Dim Side5 As System.Integer  
Dim Side6 As System.Integer  
Dim Side7 As System.Integer  
Dim Side8 As System.Integer  
Dim Side9 As System.Integer  
Dim Side10 As System.Integer  
Dim Side11 As System.Integer  
Dim Side12 As System.Integer  
Dim value As System.Boolean
```

```
value = instance.CreateSketchBelt(Length, Thickness, Engage, PulleyCount, Side1, Side2, Side3,  
Side4, Side5, Side6, Side7, Side8, Side9, Side10, Side11, Side12)
```

C#

```
System.bool CreateSketchBelt(  
    System.double Length,  
    System.double Thickness,  
    System.int Engage,  
    System.int PulleyCount,  
    System.int Side1,  
    System.int Side2,  
    System.int Side3,  
    System.int Side4,
```

```

        System.int Side5,
        System.int Side6,
        System.int Side7,
        System.int Side8,
        System.int Side9,
        System.int Side10,
        System.int Side11,
        System.int Side12
    )

```

C++/CLI

```

System.bool CreateSketchBelt(
& System.double Length,
& System.double Thickness,
& System.int Engage,
& System.int PulleyCount,
& System.int Side1,
& System.int Side2,
& System.int Side3,
& System.int Side4,
& System.int Side5,
& System.int Side6,
& System.int Side7,
& System.int Side8,
& System.int Side9,
& System.int Side10,
& System.int Side11,
& System.int Side12
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Length

Length of the belt

Thickness

Thickness of the belt

Engage

1 to engage the belt, 0 to not

PulleyCount

Number of pulley components

Side1

Place the belt for the corresponding pulley component :

- 0 = inside
- 1 = outside

Side2

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Side3

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Side4

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Side5

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Side6

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Side7

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Side8

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Side9

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Side10

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Side11

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Side12

Place the belt for the corresponding pulley component:

- 0 = inside
- 1 = outside

Return Value

True the belt is created, false if not

 Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateSketchBelt](#).

 Remarks

The order of the sides corresponds to the order of the pulleys in the selection list. For example Side1 indicates on which side of the first pulley component in the selection list you want the belt placed. If only four pulley components exist in the sketch, then Side5 - Side12 are ignored. This method limits you to 12 pulleys.

 See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

 Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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CreateSketchPlane Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateSketchPlane Method (ISketchManager)

Creates a 3D sketch plane.

 .NET Syntax

Visual Basic (Declaration)

```
Function CreateSketchPlane( _  
    ByVal Relation1 As System.Integer, _  
    ByVal Relation2 As System.Integer, _  
    ByVal Relation3 As System.Integer _  
    ) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ISketchManager  
Dim Relation1 As System.Integer
```



```
Dim Relation2 As System.Integer
Dim Relation3 As System.Integer
Dim value As System.Boolean

value = instance.CreateSketchPlane(Relation1, Relation2, Relation3)
```

C#

```
System.bool CreateSketchPlane(
    System.int Relation1,
    System.int Relation2,
    System.int Relation3
)
```

C++/CLI

```
System.bool CreateSketchPlane(
    & System.int Relation1,
    & System.int Relation2,
    & System.int Relation3
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Relation1

Relation as defined in [swConstraintType_e](#) for the first selection to position the 3D sketch plane

Relation2

Relation as defined in [swConstraintType_e](#) for the second selection to position the 3D sketch plane

Relation3

Relation as defined in [swConstraintType_e](#) for the third selection to position the 3D sketch plane

Return Value

True if the 3D sketch plane is created, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateSketchPlane](#).

Example

[Create 3D Sketch Plane \(C#\)](#)
[Create 3D Sketch Plane \(VB.NET\)](#)
[Create 3D Sketch Plane \(VBA\)](#)

Remarks

This method gets the selected items from the selection list in the order in which they were selected and applies the specified relation to them. If fewer than three items were selected, then only the first, or first and second, values are used.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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CreateSketchSlot Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateSketchSlot Method (ISketchManager)

Creates a sketch slot.

.NET Syntax

Visual Basic (Declaration)

```

Function CreateSketchSlot( _
    ByVal SlotCreationType As System.Integer, _
    ByVal SlotLengthType As System.Integer, _
    ByVal Width As System.Double, _
    ByVal X1 As System.Double, _
    ByVal Y1 As System.Double, _
    ByVal Z1 As System.Double, _
    ByVal X2 As System.Double, _
    ByVal Y2 As System.Double, _
    ByVal Z2 As System.Double, _
    ByVal X3 As System.Double, _
    ByVal Y3 As System.Double, _
    ByVal Z3 As System.Double, _
    ByVal CenterArcDirection As System.Integer, _
    ByVal AddDimension As System.Boolean _
) As SketchSlot

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim SlotCreationType As System.Integer
Dim SlotLengthType As System.Integer
Dim Width As System.Double
Dim X1 As System.Double
Dim Y1 As System.Double
Dim Z1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim Z2 As System.Double
Dim X3 As System.Double
Dim Y3 As System.Double
Dim Z3 As System.Double
Dim CenterArcDirection As System.Integer
Dim AddDimension As System.Boolean
Dim value As SketchSlot

```

```

value = instance.CreateSketchSlot(SlotCreationType, SlotLengthType, Width, X1, Y1, Z1, X2, Y2, Z2,
X3, Y3, Z3, CenterArcDirection, AddDimension)

```

C#

```

SketchSlot CreateSketchSlot(
    System.int SlotCreationType,
    System.int SlotLengthType,
    System.double Width,
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2,
    System.double X3,
    System.double Y3,
    System.double Z3,
    System.int CenterArcDirection,
    System.bool AddDimension
)

```

C++/CLI

```

SketchSlot^ CreateSketchSlot(
& System.int SlotCreationType,
& System.int SlotLengthType,
& System.double Width,
& System.double X1,
& System.double Y1,
& System.double Z1,
& System.double X2,

```

```

& System.double Y2,
& System.double Z2,
& System.double X3,
& System.double Y3,
& System.double Z3,
& System.int CenterArcDirection,
& System.bool AddDimension
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SlotCreationType

Type of sketch slot as defined in [swSketchSlotCreationType_e](#)

SlotLengthType

Type of length of sketch slot as defined in [swSketchSlotLengthType_e](#)

Width

Width of sketch slot

X1

x coordinate of point 1

Y1

y coordinate of point 1

Z1

z coordinate of point 2

X2

x coordinate of point 2

Y2

y coordinate of point 2

Z2

z coordinate of point 2

X3

x coordinate of point 3

Y3

y coordinate of point 3

Z3

z coordinate of point 3

CenterArcDirection

- 1 = Clockwise (CW)
- 1 = Counterclockwise (CCW)

AddDimension

True to automatically add dimensions, false to not

Return Value

[Sketch slot](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateSketchSlot](#).

Example

[Insert and Resize Sketch Slot \(C#\)](#)

[Insert and Resize Sketch Slot \(VB.NET\)](#)

[Insert and Resize Sketch Slot \(VBA\)](#)

See Also

[ISketchManager Interface](#)

[ISketchManager Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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CreateSpline Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateSpline Method (ISketchManager)

Obsolete. Superseded by [ISketchManager::CreateSpline2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSpline( _  
    ByVal PointData As System.Object _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim PointData As System.Object  
Dim value As SketchSegment  
  
value = instance.CreateSpline(PointData)
```

C#

```
SketchSegment CreateSpline(  
    System.object PointData  
)
```

C++/CLI

```
SketchSegment^ CreateSpline(  
& System.Object^ PointData  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointData

Array of X,Y,Z point coordinates to use in creating the spline (see **Remarks**)

Return Value

[Sketch segment](#) for the spline

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateSpline](#).

Remarks

This method creates a spline in the active 2D sketch. If a sketch is not active, then a new sketch is created. Use [ISketchManager::ActiveSketch](#) to check if the sketch active.

The PointData array is a set of, at least, two X, Y, Z values. For example, in VBA and VB.NET, the X value for the start point of the spline is PointData[0], the Y value for the start point is PointData[1], and the Z value for the start point is PointData[2]. The X value for the next point is PointData[3], and so on.

This method does not work with [ISketchManager::AddToDB](#) or [ISketchManager::DisplayWhenAdded](#). It always adds the spline directly to the database (as if ISketchManager::AddToDB(True) was in effect), and you must redraw your document window to see the entities that you added (as if ISketchManager::DisplayWhenAdded(False) was in effect).

In 2D sketches, SOLIDWORKS ignores the Z value in PointData.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateSplineByEqnParams Method](#)
[ISketchManager::CreateSplinesByEqnParams Method](#)
[ISketchManager::ICreateSpline Method](#)
[ISketchManager::ICreateSplineByEqnParams Method](#)
[ISketchManager::ICreateSplinesByEqnParams Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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CreateSpline2 Method (ISketchManager)

[See Also](#) [Example](#)



Obsolete. Superseded by [ISketchManager::CreateSpline3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSpline2( _
    ByVal PointData As System.Object, _
    ByVal SimulateNaturalEnds As System.Boolean _
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim PointData As System.Object
Dim SimulateNaturalEnds As System.Boolean
Dim value As SketchSegment

value = instance.CreateSpline2(PointData, SimulateNaturalEnds)
```

C#

```
SketchSegment CreateSpline2(
    System.object PointData,
    System.bool SimulateNaturalEnds
)
```

C++/CLI

```
SketchSegment^ CreateSpline2(
    & System.Object^ PointData,
    & System.bool SimulateNaturalEnds
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointData

Array of X,Y,Z point coordinates to use in creating the spline (see **Remarks**)

SimulateNaturalEnds

True to simulate natural ends, false to not simulate natural ends

Return Value

[Sketch segment](#) for the spline

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateSpline2](#).

Example

[Insert a Composite Curve \(C#\)](#)
[Insert a Composite Curve \(VB.NET\)](#)
[Insert a Composite Curve \(VBA\)](#)

Remarks

This method creates a spline in the active 2D sketch. If a sketch is not active, then a new sketch is created. Use [ISketchManager::ActiveSketch](#) to check if the sketch active.

The PointData array is a set of, at least, two X, Y, Z values. For example, in VBA and VB.NET, the X value for the start point of the spline is PointData[0], the Y value for the start point is PointData[1], and the Z value for the start point is PointData[2]. The X value for the next point is PointData[3], and so on.

This method does not work with [ISketchManager::AddToDB](#) or [ISketchManager::DisplayWhenAdded](#). It always adds the spline directly to the database (as if ISketchManager::AddToDB(True) was in effect), and you must redraw your document window to see the entities that you added (as if ISketchManager::DisplayWhenAdded(False) was in effect).

In 2D sketches, SOLIDWORKS ignores the Z value in PointData.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::ICreateSpline2 Method](#)
[ISketchManager::CreateSplineByEqnParams Method](#)
[ISketchManager::CreateSplinesByEqnParams Method](#)
[ISketchManager::ICreateSplineByEqnParams Method](#)
[ISketchManager::ICreateSplinesByEqnParams Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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CreateSpline3 Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateSpline3 Method (ISketchManager)

Creates either a 2D spline or a spline constrained to a surface.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSpline3( _  
    ByVal PointData As System.Object, _  
    ByVal Surfs As System.Object, _  
    ByVal Direction As System.Object, _  
    ByVal SimulateNaturalEnds As System.Boolean, _  
    ByRef Status As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim PointData As System.Object  
Dim Surfs As System.Object  
Dim Direction As System.Object  
Dim SimulateNaturalEnds As System.Boolean  
Dim Status As System.Object  
Dim value As System.Object  
  
value = instance.CreateSpline3(PointData, Surfs, Direction, SimulateNaturalEnds, Status)
```

C#

```
System.object CreateSpline3(  
    System.object PointData,  
    System.object Surfs,  
    System.object Direction,  
    System.bool SimulateNaturalEnds,  
    out System.object Status  
)
```

C++/CLI

```
System.Object^ CreateSpline3(  
& System.Object^ PointData,  
& System.Object^ Surfs,  
& System.Object^ Direction,  
& System.bool SimulateNaturalEnds,  
& [Out] System.Object^ Status  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointData

Array of X,Y,Z coordinates of the spline points (see **Remarks**)

Surfs

Array of [ISurface](#) objects; null or Nothing for 2D splines (see **Remarks**)
Direction
Array of [IMathVector](#) objects; valid only for on-surface splines (see **Remarks**)
SimulateNaturalEnds
True to simulate natural ends, false to not; valid only for 2D splines (see **Remarks**)
Status
Array of boolean values; empty for 2D splines (see **Remarks**)

Return Value

[ISketchSegment](#) or null if an error

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateSpline3](#).

Example

[Create On-surface Spline \(VBA\)](#)
[Create On-surface Spline \(VB.NET\)](#)
[Create On-surface Spline \(C#\)](#)

Remarks

This method allows you to create:

- 2D splines.
- On-surface splines as can also be created using **Tools > Sketch Entities > Spline on Surface**.

To create a 2D spline, specify:

- PointData with at least six coordinates (for the start and end points of the spline):

`[start_pt_x, start_pt_y, start_pt_z, end_pt_x, end_pt_y, end_pt_z]`

For 2D sketches, SOLIDWORKS assumes PointData contains no Z values.

- SimulateNaturalEnds with true for zero curvature end conditions or false to maintain curvature at the ends.
- All other parameters with null or Nothing.

To create an on-surface spline, specify:

- PointData with two or more points anywhere in 3D space. Each spline point will project onto a surface in the Surf array, depending on the direction vector specified for each point in the Direction array.
- Surf with an array of surfaces onto which the PointData is projected.
- Direction with an array of vectors, which are projection directions for every point in the PointData array. If Direction is Nothing or null, then the view vector for the current screen view used to project each point onto a surface.

For on-surface splines only, Status contains an array of statuses, one for each point in PointArray. True indicates that the point successfully projects onto one of the surfaces. False indicates either:

- the point does not "hit" any of the specified surfaces using the direction that the point is mapped to in the Direction array,
- OR -
- the spline cannot be continued at that point where the surfaces are no longer contiguous.

This method:

- Creates the 2D spline in the active sketch. If a sketch is not active, then a new sketch is created. Use [ISketchManager::ActiveSketch](#) to check if the sketch active.
- Creates a new 3D sketch to create the on-surface spline. Call [ISketchManager::InsertSketch](#) to finalize the sketch.
- Does not work with [ISketchManager::AddToDB](#) or [ISketchManager::DisplayWhenAdded](#). It always adds the spline directly to the database (as if [ISketchManager::AddToDB\(True\)](#) was in effect), and you must redraw your document window to see the entities that you added (as if [ISketchManager::DisplayWhenAdded\(False\)](#) was in effect).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateSplineByEqnParams Method \(\)](#)
[ISketchManager::CreateSplinesByEqnParams2 Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateSplineByEqnParams Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateSplineByEqnParams Method (ISketchManager)

Creates a B-curve from B-spline data; that is, a set of B-spline vertices (control points) and a knot vector.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSplineByEqnParams( _
    ByVal Parameters As System.Object _
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim Parameters As System.Object
Dim value As SketchSegment

value = instance.CreateSplineByEqnParams(Parameters)
```

C#

```
SketchSegment CreateSplineByEqnParams(
    System.object Parameters
)
```

C++/CLI

```
SketchSegment^ CreateSplineByEqnParams(
    & System.Object^ Parameters
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Parameters

Array containing an array of doubles to use in creating the spline (see **Remarks**)

Return Value

[Sketch segment](#) for the spline

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateSplineByEqnParams](#).

Remarks

The Properties array contains four longs (placed in the first four doubles in the array):

- Dimension
- Order
- Number of Control Points
- Periodicity (true or false)

The Knots array contains doubles with (Number of Control Points + Order) elements.

The size of the ControlPoints array is based upon the curve dimension.

- Dimension = 2 then each Control Point is an array of 2 doubles (x, y)
- Dimension = 3 then each Control Point is an array of 3 doubles (x, y, z)
- Dimension = 4 then each Control Point is an array of 4 doubles (x, y, z, w) where w = weight

The parameters are provided as three arrays.

Pass the control point coordinates to this routine in sketch space. The Z value is interpreted as 0. In certain situations, you must transform your B-curve parameters to sketch space using [ISketch::ModelToSketchTransform](#).

NOTE: The curve can be either periodic or non-periodic. If the generated spline is a closed spline, then it must be flagged as periodic. Additionally, splines generated in sketches must be G1 continuous. If the data passed to this method does not generate a G1 continuous spline, then it is rejected and a spline is not created. If your data is not G1 continuous, then you must split the spline into multiple G1 segments and call this method for each segment.

This method does not work with [ISketchManager::AddToDB](#) or [ISketchManager::DisplayWhenAdded](#). It always adds the spline directly to the database (as if [ISketchManager::AddToDB\(True\)](#) is in effect), and you must redraw your document window to see the entities that you added (as if [ISketchManager::DisplayWhenAdded\(False\)](#) is in effect).

To create 3D splines, see [IModelDoc2::InsertCurveFilePoint](#) and related functions.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateSpline Method](#)
[ISketchManager::CreateSplinesByEqnParams Method](#)
[ISketchManager::ICreateSpline Method](#)
[ISketchManager::ICreateSplineByEqnParams Method](#)
[ISketchManager::ICreateSplinesByEqnParams Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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CreateSplineParamData Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateSplineParamData Method (ISketchManager)

Creates an empty spline parameter data object.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSplineParamData() As SplineParamData
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim value As SplineParamData  
  
value = instance.CreateSplineParamData()
```

C#

```
SplineParamData CreateSplineParamData()
```

C++/CLI

```
SplineParamData^ CreateSplineParamData();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[ISplineParamData](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateSplineParamData](#).

Example

[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateSplinesByEqnParams2 Method](#)
[ISketch::GetSplineParams4 Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateSplinesByEqnParams Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateSplinesByEqnParams Method (ISketchManager)

Obsolete. Superseded by [ISketchManager::CreateSplinesByEqnParams2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSplinesByEqnParams( _  
    ByVal Parameters As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim Parameters As System.Object  
Dim value As System.Object
```

```
value = instance.CreateSplinesByEqnParams(Parameters)
```

C#

```
System.object CreateSplinesByEqnParams(  
    System.object Parameters  
)
```

C++/CLI

```
System.Object^ CreateSplinesByEqnParams(  
& System.Object^ Parameters  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Parameters

Array containing an array of doubles to use in creating the spline (see **Remarks**)

Return Value

Array of [sketch segments](#) of the newly created splines

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateSplinesByEqnParams](#).

Remarks

The Properties array contains four longs (placed in the first four doubles in the array):

- Dimension
- Order
- Number of Control Points
- Periodicity (true or false)

The Knots array contains doubles with (Number of Control Points + Order) elements.

The size of the ControlPoints array is based upon the curve dimension.

- Dimension = 2 then each Control Point is an array of 2 doubles (x, y)
- Dimension = 3 then each Control Point is an array of 3 doubles (x, y, z)
- Dimension = 4 then each Control Point is an array of 4 doubles (x, y, z, w) where w = weight

The parameters are provided as three arrays.

Pass the control point coordinates to this routine in sketch space. The Z value is interpreted as 0. In certain situations, you must transform your B-curve parameters to sketch space using [ISketch::ModelToSketchTransform](#).

NOTE: If the spline being generated is a closed spline, then it must be flagged as periodic. If the data passed to this method does not generate a G1 continuous spline, then it creates multiple G1 continuous spline segments.

This method does not work with [ISketchManager::AddToDB](#) or [ISketchManager::DisplayWhenAdded](#). It always adds the spline directly to the database (as if [ISketchManager::AddToDB\(True\)](#) is in effect), and you must redraw your document window to see the entities that you added (as if [ISketchManager::DisplayWhenAdded\(False\)](#) is in effect).

To create 3D splines, see [IModelDoc2::InsertCurveFilePoint](#) and related functions.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateSpline Method](#)
[ISketchManager::CreateSplineByEqnParams Method](#)
[ISketchManager::ICreateSpline Method](#)
[ISketchManager::ICreateSplineByEqnParams Method](#)
[ISketchManager::ICreateSplinesByEqnParams Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateSplinesByEqnParams2 Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateSplinesByEqnParams2 Method (ISketchManager)

Creates one or more spline segments using the B-curve parameters provided.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateSplinesByEqnParams2( _  
    ByVal ParamData As SplineParamData _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim ParamData As SplineParamData  
Dim value As System.Object  
  
value = instance.CreateSplinesByEqnParams2(ParamData)
```

C#

```
System.Object CreateSplinesByEqnParams2(  
    SplineParamData ParamData  
)
```

C++/CLI

```
System.Object^ CreateSplinesByEqnParams2(  
& SplineParamData^ ParamData  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ParamData
[ISplineParamData](#)

Return Value

Array of [sketch segments](#) of the newly created spline

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateSplinesByEqnParams2](#).

Example

[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateSpline2 Method](#)
[ISketchManager::CreateSplineByEqnParams Method](#)
[ISketchManager::ICreateSpline2 Method](#)
[ISketchManager::ICreateSplineByEqnParams Method](#)
[ISketchManager::ICreateSplinesByEqnParams Method](#)
[ISketchManager::CreateSplineParamData Method](#)
[ISketch::GetSplineParams4 Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateTangentArc Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateTangentArc Method (ISketchManager)

Creates a tangent arc.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateTangentArc( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double, _  
    ByVal ArcType As System.Integer _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim Z2 As System.Double  
Dim ArcType As System.Integer  
Dim value As SketchSegment  
  
value = instance.CreateTangentArc(X1, Y1, Z1, X2, Y2, Z2, ArcType)
```

C#

```
SketchSegment CreateTangentArc(  
    System.double X1,  
    System.double Y1,  
    System.double Z1,  
    System.double X2,  
    System.double Y2,  
    System.double Z2,  
    System.int ArcType  
)
```

C++/CLI

```
SketchSegment^ CreateTangentArc(  
& System.double X1,  
& System.double Y1,  
& System.double Z1,  
& System.double X2,  
& System.double Y2,  
& System.double Z2,  
& System.int ArcType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X1](#)
X coordinate of start point in meters
[Y1](#)
Y coordinate of start point in meters
[Z1](#)
Z coordinate of start point in meters
[X2](#)
X coordinate of end point in meters
[Y2](#)
Y coordinate of end point in meters
[Z2](#)
Z coordinate of end point in meters
[ArcType](#)
Type of tangent arc as defined in [swTangentArcTypes](#) e

Return Value

[Sketch segment](#) of the tangent arc

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateTangentArc](#).

Example

[Create Tangent Arc \(VBA\)](#)
[Create Tangent Arc \(VB.NET\)](#)
[Create Tangent Arc \(C#\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

CurvatureDensity Property (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CurvatureDensity Property (ISketchManager)

Gets or sets the scaling factor by which to adjust the density of the curvature combs for this spline.

.NET Syntax

Visual Basic (Declaration)

Property CurvatureDensity As System.Integer

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim value As System.Integer

instance.CurvatureDensity = value

value = instance.CurvatureDensity
```

C#

```
System.int CurvatureDensity {get; set;}
```

C++/CLI

```
property System.int CurvatureDensity {
    System.int get();
    void set ( &    System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Curvature density of the curvature combs for this spline; valid values from 0 - 100

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CurvatureDensity](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CurvatureScale Property](#)
[ISketchSpline::ShowCurvatureCombs Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

CurvatureScale Property (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CurvatureScale Property (ISketchManager)

Gets or sets the scaling factor by which to adjust the size of the curvature combs for this spline.

.NET Syntax

Visual Basic (Declaration)

```
Property CurvatureScale As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim value As System.Double

instance.CurvatureScale = value

value = instance.CurvatureScale
```

C#

```
System.double CurvatureScale {get; set;}
```

C++/CLI

```
property System.double CurvatureScale {
    System.double get();
```

```
void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Curvature density of the curvature combs for this spline; valid values from 0 - 100

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CurvatureScale](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchSpline::ShowCurvatureCombs Property](#)
[ISketchManager::CurvatureDensity Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

DisplayWhenAdded Property (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : DisplayWhenAdded Property (ISketchManager)

Gets or sets whether new sketch entities are immediately displayed when created.

.NET Syntax

Visual Basic (Declaration)

```
Property DisplayWhenAdded As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim value As System.Boolean  
  
instance.DisplayWhenAdded = value  
  
value = instance.DisplayWhenAdded
```

C#

```
System.bool DisplayWhenAdded {get; set;}
```

C++/CLI

```
property System.bool DisplayWhenAdded {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to display new sketch entities when added, false to not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::DisplayWhenAdded](#).

Remarks

The sketch entities appear on the screen after performing [IModelView::GraphicsRedraw](#) or [IModelView::IGraphicsRedraw](#) or [IModelDoc2::EditRebuild3](#) is performed. Also, [ISketchManager::AddToDB](#) must be true to use this method's settings.

This display setting remains even after your program run has ended. Therefore, it is recommended that you reset this parameter to TRUE at the end of your program. For example, if you have ended your program and the display is set to FALSE, then the user would have difficulty performing selections and newly created entities would not be seen until a redraw or a rebuild.

NOTES:

- IModelView::GraphicsRedraw and IModelView::IGraphicsRedraw are much faster than IModelDoc2::EditRebuild3.
- ISketchManager::AddToDB and ISketchManager::DisplayWhenAdded also increase performance during sketch entity creation.

 See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

 Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Document Property (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : Document Property (ISketchManager)

Gets the document for this [ISketchManager](#) object.

 .NET Syntax

Visual Basic (Declaration)

ReadOnly Property Document As ModelDoc2

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim value As ModelDoc2  
  
value = instance.Document
```

C#

```
ModelDoc2 Document {get;}
```

C++/CLI

```
property ModelDoc2^ Document {  
    ModelDoc2^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Document](#)

 Visual Basic for Applications (VBA) Syntax

See [SketchManager::Document](#).

 See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

 Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EditCircularSketchStepAndRepeat Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : EditCircularSketchStepAndRepeat Method (ISketchManager)

Edits a circular sketch pattern.

.NET Syntax

Visual Basic (Declaration)

```
Function EditCircularSketchStepAndRepeat( _  
    ByVal ArcRadius As System.Double, _  
    ByVal ArcAngle As System.Double, _  
    ByVal PatternNum As System.Integer, _  
    ByVal PatternSpacing As System.Double, _  
    ByVal PatternRotate As System.Boolean, _  
    ByVal DeleteInstances As System.String, _  
    ByVal RadiusDim As System.Boolean, _  
    ByVal AngleDim As System.Boolean, _  
    ByVal CreateNumOfInstancesDim As System.Boolean, _  
    ByVal Seeds As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim ArcRadius As System.Double  
Dim ArcAngle As System.Double  
Dim PatternNum As System.Integer  
Dim PatternSpacing As System.Double  
Dim PatternRotate As System.Boolean  
Dim DeleteInstances As System.String  
Dim RadiusDim As System.Boolean  
Dim AngleDim As System.Boolean  
Dim CreateNumOfInstancesDim As System.Boolean  
Dim Seeds As System.String  
Dim value As System.Boolean
```

```
value = instance.EditCircularSketchStepAndRepeat(ArcRadius, ArcAngle, PatternNum, PatternSpacing,  
PatternRotate, DeleteInstances, RadiusDim, AngleDim, CreateNumOfInstancesDim, Seeds)
```

C#

```
System.bool EditCircularSketchStepAndRepeat(  
    System.double ArcRadius,  
    System.double ArcAngle,  
    System.int PatternNum,  
    System.double PatternSpacing,  
    System.bool PatternRotate,  
    System.string DeleteInstances,  
    System.bool RadiusDim,  
    System.bool AngleDim,  
    System.bool CreateNumOfInstancesDim,  
    System.string Seeds  
)
```

C++/CLI

```
System.bool EditCircularSketchStepAndRepeat(  
& System.double ArcRadius,  
& System.double ArcAngle,  
& System.int PatternNum,  
& System.double PatternSpacing,  
& System.bool PatternRotate,  
& System.String^ DeleteInstances,  
& System.bool RadiusDim,  
& System.bool AngleDim,  
& System.bool CreateNumOfInstancesDim,
```

```
& System.String^ Seeds  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArcRadius

Radius of the circular sketch pattern

ArcAngle

Angle relative to the sketch entities being patterned

PatternNum

Total number of instances, including the seed geometry

PatternSpacing

Spacing between pattern instances

PatternRotate

True to rotate the pattern, false to not

DeleteInstances

Number of instances to delete, passed as a string formatted as: "(a) (b) (c) "

RadiusDim

True to display the radius dimension in the graphics area, false to not

AngleDim

True to display the angle dimension in the graphics area, false to not

CreateNumOfInstancesDim

True to display the number of instances dimension in the graphics area, false to not

Seeds

Array of the names of the entities, separated by the underscore character (_), that comprise the seed pattern (e.g., "Line1_Line2_Line3_Line4" for a rectangular-shaped seed pattern)

Return Value

True if the circular sketch pattern is successfully edited, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::EditCircularSketchStepAndRepeat](#).

Example

[Create and Edit Circular Sketch Pattern \(VB.NET\)](#)

[Create and Edit Circular Sketch Pattern \(VBA\)](#)

[Create and Edit Circular Sketch Pattern \(C#\)](#)

Remarks

The radius, angle, and number of instances dimensions displayed in the graphics area can be modified by interactive users.

See Also

[ISketchManager Interface](#)

[ISketchManager Members](#)

[ISketchManager::CreateCircularSketchStepAndRepeat Method](#)

[ISketchManager::CreateLinearSketchStepAndRepeat Method](#)

[ISketchManager::EditLinearSketchStepAndRepeat Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EditLinearSketchStepAndRepeat Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : EditLinearSketchStepAndRepeat Method (ISketchManager)

Edits a linear sketch pattern.

.NET Syntax

Visual Basic (Declaration)

```
Function EditLinearSketchStepAndRepeat( _  
    ByVal NumX As System.Integer, _  
    ByVal NumY As System.Integer, _  
    ByVal SpacingX As System.Double, _  
    ByVal SpacingY As System.Double, _  
    ByVal AngleX As System.Double, _
```

```

    ByVal AngleY As System.Double, _
    ByVal DeleteInstances As System.String, _
    ByVal XSpacingDim As System.Boolean, _
    ByVal YSpacingDim As System.Boolean, _
    ByVal AngleDim As System.Boolean, _
    ByVal CreateNumOfInstancesDimInXDir As System.Boolean, _
    ByVal CreateNumOfInstancesDimInYDir As System.Boolean, _
    ByVal Seeds As System.String _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim NumX As System.Integer
Dim NumY As System.Integer
Dim SpacingX As System.Double
Dim SpacingY As System.Double
Dim AngleX As System.Double
Dim AngleY As System.Double
Dim DeleteInstances As System.String
Dim XSpacingDim As System.Boolean
Dim YSpacingDim As System.Boolean
Dim AngleDim As System.Boolean
Dim CreateNumOfInstancesDimInXDir As System.Boolean
Dim CreateNumOfInstancesDimInYDir As System.Boolean
Dim Seeds As System.String
Dim value As System.Boolean

```

```

value = instance.EditLinearSketchStepAndRepeat(NumX, NumY, SpacingX, SpacingY, AngleX, AngleY,
DeleteInstances, XSpacingDim, YSpacingDim, AngleDim, CreateNumOfInstancesDimInXDir,
CreateNumOfInstancesDimInYDir, Seeds)

```

C#

```

System.bool EditLinearSketchStepAndRepeat(
    System.int NumX,
    System.int NumY,
    System.double SpacingX,
    System.double SpacingY,
    System.double AngleX,
    System.double AngleY,
    System.string DeleteInstances,
    System.bool XSpacingDim,
    System.bool YSpacingDim,
    System.bool AngleDim,
    System.bool CreateNumOfInstancesDimInXDir,
    System.bool CreateNumOfInstancesDimInYDir,
    System.string Seeds
)

```

C++/CLI

```

System.bool EditLinearSketchStepAndRepeat(
    & System.int NumX,
    & System.int NumY,
    & System.double SpacingX,
    & System.double SpacingY,
    & System.double AngleX,
    & System.double AngleY,
    & System.String^ DeleteInstances,
    & System.bool XSpacingDim,
    & System.bool YSpacingDim,
    & System.bool AngleDim,
    & System.bool CreateNumOfInstancesDimInXDir,
    & System.bool CreateNumOfInstancesDimInYDir,
    & System.String^ Seeds
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumX
Total number of instances along the x axis, including the seed

NumY
Total number of instances along the y axis, including the seed
SpacingX
Spacing between instances along the x axis
SpacingY
Spacing between instances along the y axis
AngleX
Angle for direction 1 relative to the x axis

AngleY
Angle for direction 2 relative to the y axis

DeleteInstances
Number of instances to delete, passed as a string in the format: "(a) (b) (c) "

XSpacingDim
True to display the spacing between instances dimension along the x axis in the graphics area, false to not
YSpacingDim
True to display the spacing between instances dimension along the y axis in the graphics area, false to not
AngleDim
True to display the angle dimension between axes in the graphics area, false to not

CreateNumOfInstancesDimInXDir
True to display the number of instances in the x direction dimension in the graphics area, false to not
CreateNumOfInstancesDimInYDir
True to display the number of instances in the y direction dimension in the graphics area, false to not
Seeds
Array of the names of the entities, separated by the underscore character (_), that comprise the seed pattern (e.g., Line1_Line2_Line3_Line4 for a rectangular-shaped seed pattern)

Return Value

True if the linear sketch pattern is edited, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::EditLinearSketchStepAndRepeat](#).

Example

[Create and Edit Linear Sketch Pattern \(VB.NET\)](#)
[Create and Edit Linear Sketch Pattern \(VBA\)](#)
[Create and Edit Linear Sketch Pattern \(C#\)](#)

Remarks

The spacing instances, angle, and number of instances dimensions displayed in the graphics area can be modified by interactive users

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateLinearSketchStepAndRepeat Method](#)
[ISketchManager::CreateCircularSketchStepAndRepeat Method](#)
[ISketchManager::EditCircularSketchStepAndRepeat Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

EditSketchBlock Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : EditSketchBlock Method (ISketchManager)

Puts the block definition in edit mode.

.NET Syntax

Visual Basic (Declaration)

Sub EditSketchBlock()

Visual Basic (Usage)



```
Dim instance As ISketchManager
```

```
instance.EditSketchBlock()
```

C#

```
void EditSketchBlock()
```

C++/CLI

```
void EditSketchBlock();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SketchManager::EditSketchBlock](#).

Remarks

When done editing the block definition, use [ISketchManager::EndEditSketchBlock](#).

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchBlockDefinition Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EndEditSketchBlock Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : EndEditSketchBlock Method (ISketchManager)

Saves or discards your edits of the block and then ends the current editing session of this block.

.NET Syntax

Visual Basic (Declaration)

```
Sub EndEditSketchBlock( _  
    ByVal AcceptChanges As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim AcceptChanges As System.Boolean  
  
instance.EndEditSketchBlock(AcceptChanges)
```

C#

```
void EndEditSketchBlock(  
    System.bool AcceptChanges  
)
```

C++/CLI

```
void EndEditSketchBlock(  
& System.bool AcceptChanges
```

)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AcceptChanges

True to save your edits, false to discard them

Visual Basic for Applications (VBA) Syntax

See [SketchManager::EndEditSketchBlock](#).

Remarks

Use [ISketchManager::EditSketchBlock](#) to put the block in edit mode.

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchBlockDefinition Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ExplodeSketchBlockInstance Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : ExplodeSketchBlockInstance Method (ISketchManager)

Explodes the specified block instance.

.NET Syntax

Visual Basic (Declaration)

```
Sub ExplodeSketchBlockInstance( _  
    ByVal LpSketchBlockInstance As SketchBlockInstance _  
)
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim LpSketchBlockInstance As SketchBlockInstance  
  
instance.ExplodeSketchBlockInstance(LpSketchBlockInstance)
```

C#

```
void ExplodeSketchBlockInstance(  
    SketchBlockInstance LpSketchBlockInstance  
)
```

C++/CLI

```
void ExplodeSketchBlockInstance(  
& SketchBlockInstance^ LpSketchBlockInstance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

LpSketchBlockInstance
[Block instance](#) to explode

Visual Basic for Applications (VBA) Syntax

See [SketchManager::ExplodeSketchBlockInstance](#).

Remarks

If this is the only or last block instance, then:

- All sketch entities in the block instances are copied to the owning sketch of the block instance.
- Source block definition is destroyed.

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchBlockInstance Interface](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FullyDefineSketch Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : FullyDefineSketch Method (ISketchManager)

Fully defines a sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function FullyDefineSketch( _  
    ByVal EntitiesToFullyDefine As System.Boolean, _  
    ByVal UseRelations As System.Boolean, _  
    ByVal RelationsToApply As System.Integer, _  
    ByVal UseDimensions As System.Boolean, _  
    ByVal HorizontalDimScheme As System.Integer, _  
    ByVal HorizontalDatumDisp As System.Object, _  
    ByVal VerticalDimScheme As System.Integer, _  
    ByVal VerticalDatumDisp As System.Object, _  
    ByVal HorizontalDimPlacement As System.Integer, _  
    ByVal VerticalDimPlacement As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim EntitiesToFullyDefine As System.Boolean  
Dim UseRelations As System.Boolean  
Dim RelationsToApply As System.Integer  
Dim UseDimensions As System.Boolean  
Dim HorizontalDimScheme As System.Integer  
Dim HorizontalDatumDisp As System.Object  
Dim VerticalDimScheme As System.Integer  
Dim VerticalDatumDisp As System.Object  
Dim HorizontalDimPlacement As System.Integer  
Dim VerticalDimPlacement As System.Integer  
Dim value As System.Integer  
  
value = instance.FullyDefineSketch(EntitiesToFullyDefine, UseRelations, RelationsToApply,  
    UseDimensions, HorizontalDimScheme, HorizontalDatumDisp, VerticalDimScheme, VerticalDatumDisp,  
    HorizontalDimPlacement, VerticalDimPlacement)
```

C#

```

System.int FullyDefineSketch(
    System.bool EntitiesToFullyDefine,
    System.bool UseRelations,
    System.int RelationsToApply,
    System.bool UseDimensions,
    System.int HorizontalDimScheme,
    System.object HorizontalDatumDisp,
    System.int VerticalDimScheme,
    System.object VerticalDatumDisp,
    System.int HorizontalDimPlacement,
    System.int VerticalDimPlacement
)

```

C++/CLI

```

System.int FullyDefineSketch(
    & System.bool EntitiesToFullyDefine,
    & System.bool UseRelations,
    & System.int RelationsToApply,
    & System.bool UseDimensions,
    & System.int HorizontalDimScheme,
    & System.Object^ HorizontalDatumDisp,
    & System.int VerticalDimScheme,
    & System.Object^ VerticalDatumDisp,
    & System.int HorizontalDimPlacement,
    & System.int VerticalDimPlacement
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

EntitiesToFullyDefine

True to fully define all entities, false to fully define only the entities selected

UseRelations

True to use relations, false to not

RelationsToApply

Relations to apply as defined in [swSketchFullyDefineRelationType_e](#)

UseDimensions

True to use dimensions, false to not

HorizontalDimScheme

- 0 = Chain

- 1 = Baseline

- 2 = Ordinate

HorizontalDatumDisp

Horizontal datum (model edge, model vertex, sketch line, sketch point), or, if Nothing or null, use entity with selection mark = 6

VerticalDimScheme

- 0 = Chain

- 1 = Baseline

- 2 = Ordinate

VerticalDatumDisp

Vertical datum (model edge, model vertex, sketch line, sketch point), or, if Nothing or null, use entity with selection mark = 6

HorizontalDimPlacement

- 0 = Above sketch

- 1 = Below sketch

VerticalDimPlacement

- 0 = Right of sketch

- 1 = Left of sketch

Return Value

Not currently defined

Visual Basic for Applications (VBA) Syntax

See [SketchManager::FullyDefineSketch](#).

Example

[Fully Define Under Defined Sketch \(VBA\)](#)

[Fully Define Under Defined Sketch \(VB.NET\)](#)

[Fully Define Under Defined Sketch \(C#\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::AddAlongXDimension Method](#)
[ISketchManager::AddAlongYDimension Method](#)
[ISketchManager::AddAlongZDimension Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetDynamicMirror Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : GetDynamicMirror Method (ISketchManager)

Gets whether dynamic sketch mirroring, which is the automatic mirroring of newly created sketch entities about a selected centerline, is enabled.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDynamicMirror() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim value As System.Boolean  
  
value = instance.GetDynamicMirror()
```

C#

```
System.bool GetDynamicMirror()
```

C++/CLI

```
System.bool GetDynamicMirror();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if dynamic sketch mirroring is enabled, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::GetDynamicMirror](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::SetDynamicMirror Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSketchBlockDefinitionCount Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : GetSketchBlockDefinitionCount Method (ISketchManager)

Gets the number of block definitions in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchBlockDefinitionCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim value As System.Integer

value = instance.GetSketchBlockDefinitionCount()
```

C#

```
System.int GetSketchBlockDefinitionCount()
```

C++/CLI

```
System.int GetSketchBlockDefinitionCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of block definitions in the model

Visual Basic for Applications (VBA) Syntax

See [SketchManager::GetSketchBlockDefinitionCount](#).

Remarks

Call this method before calling [ISketchManager::IGetSketchBlockDefinitions](#) to get the size of the array for that method.

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::GetSketchBlockDefinitions Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSketchBlockDefinitions Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : GetSketchBlockDefinitions Method (ISketchManager)

Gets all of the block definitions.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketchBlockDefinitions() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim value As System.Object

value = instance.GetSketchBlockDefinitions()
```

C#

```
System.object GetSketchBlockDefinitions()
```

C++/CLI

```
System.Object^ GetSketchBlockDefinitions();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Block definitions](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::GetSketchBlockDefinitions](#).

Example

[Get and Set Blocks Data in Any Document \(VBA\)](#)
[Get Block Definitions in Part or Assembly \(VBA\)](#)
[Get Block Information \(VBA\)](#)

Remarks

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::GetSketchBlockDefinitionCount Method](#)
[ISketchManager::IGetSketchBlockDefinitions Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateSpline Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : ICreateSpline Method (ISketchManager)

Obsolete. Superseded by [ISketchManager::ICreateSpline2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSpline( _  
    ByVal PointCount As System.Integer, _  
    ByRef PointData As System.Double _  
) As SketchSegment
```

Visual Basic (Usage)

```
Dim instance As ISketchManager  
Dim PointCount As System.Integer  
Dim PointData As System.Double  
Dim value As SketchSegment
```

```
value = instance.ICreateSpline(PointCount, PointData)
```

C#

```
SketchSegment ICreateSpline(  
    System.int PointCount,  
    ref System.double PointData  
)
```

C++/CLI



```
SketchSegment^ ICreateSpline(  
& System.int PointCount,  
& System.double% PointData  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointCount

Number of points in the spline

PointData

Array of X,Y,Z point coordinates to use in creating the spline (see **Remarks**)

Return Value

[Sketch segment](#) for the spline

Visual Basic for Applications (VBA) Syntax

See [SketchManager::ICreateSpline](#).

Remarks

This method creates a spline in the active 2D sketch. If a sketch is not active, then a new sketch is created. Use [ISketchManager::ActiveSketch](#) to check if the sketch active.

The *PointData* array is a set of, at least, two X, Y, Z values. The X value for the start point of the spline is *PointData*[0], the Y value for the start point is *PointData*[1], and the Z value for the start point is *PointData*[2]. The X value for the next point is *PointData*[3], and so on.

This method does not work with [ISketchManager::AddToDB](#) or [ISketchManager::DisplayWhenAdded](#). It always adds the spline directly to the database (as if *ISketchManager::AddToDB*(True) was in effect), and you must redraw your document window to see the entities that you added (as if *ISketchManager::DisplayWhenAdded*(False) was in effect).

In 2D sketches, SOLIDWORKS ignores the Z value in *PointData*.

See Also

[ISketchManager Interface](#)

[ISketchManager Members](#)

[ISketchManager::CreateSpline Method](#)

[ISketchManager::CreateSplineByEqnParams Method](#)

[ISketchManager::CreateSplinesByEqnParams Method](#)

[ISketchManager::ICreateSplineByEqnParams Method](#)

[ISketchManager::ICreateSplinesByEqnParams Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ICreateSpline2 Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : ICreateSpline2 Method (ISketchManager)

Creates a spline passing through the given points.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSpline2( _  
    ByVal PointCount As System.Integer, _  
    ByRef PointData As System.Double, _  
    ByVal SimulateNaturalEnds As System.Boolean _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim PointCount As System.Integer  
Dim PointData As System.Double
```

```
Dim SimulateNaturalEnds As System.Boolean
Dim value As SketchSegment

value = instance.ICreateSpline2(PointCount, PointData, SimulateNaturalEnds)
```

C#

```
SketchSegment ICreateSpline2(
    System.int PointCount,
    ref System.double PointData,
    System.bool SimulateNaturalEnds
)
```

C++/CLI

```
SketchSegment^ ICreateSpline2(
    & System.int PointCount,
    & System.double% PointData,
    & System.bool SimulateNaturalEnds
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointCount

Number of points in the spline

PointData

- In-process, unmanaged C++: Pointer to an array of X,Y,Z point coordinates to use in creating the spline (see **Remarks**)

• VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

SimulateNaturalEnds

True to simulate natural ends, false to not

Return Value

[Sketch segment](#) for the spline

Remarks

This method creates a spline in the active 2D sketch. If a sketch is not active, then a new sketch is created. Use [ISketchManager::ActiveSketch](#) to check if the sketch active.

The PointData array is a set of, at least, two X, Y, Z values. The X value for the start point of the spline is PointData[0], the Y value for the start point is PointData[1], and the Z value for the start point is PointData[2]. The X value for the next point is PointData[3], and so on.

This method does not work with [ISketchManager::AddToDB](#) or [ISketchManager::DisplayWhenAdded](#). It always adds the spline directly to the database (as if ISketchManager::AddToDB(True) was in effect), and you must redraw your document window to see the entities that you added (as if ISketchManager::DisplayWhenAdded(False) was in effect).

In 2D sketches, SOLIDWORKS ignores the Z value in PointData.

See Also

[ISketchManager Interface](#)

[ISketchManager Members](#)

[ISketchManager::CreateSpline2 Method](#)

[ISketchManager::CreateSplineByEqnParams Method](#)

[ISketchManager::CreateSplinesByEqnParams Method](#)

[ISketchManager::ICreateSplineByEqnParams Method](#)

[ISketchManager::ICreateSplinesByEqnParams Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateSplineByEqnParams Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : ICreateSplineByEqnParams Method (ISketchManager)

Creates a B-curve from B-spline data; that is, a set of B-spline vertices (control points) and a knot vector.

.NET Syntax

```

Function ICreateSplineByEqnParams( _
    ByRef Properties As System.Integer, _
    ByVal KnotArrayCount As System.Integer, _
    ByRef Knots As System.Double, _
    ByVal ControlPointArrayCount As System.Integer, _
    ByRef ControlPoints As System.Double _
) As SketchSegment

```

```

Dim instance As ISketchManager
Dim Properties As System.Integer
Dim KnotArrayCount As System.Integer
Dim Knots As System.Double
Dim ControlPointArrayCount As System.Integer
Dim ControlPoints As System.Double
Dim value As SketchSegment

value = instance.ICreateSplineByEqnParams(Properties, KnotArrayCount, Knots,
ControlPointArrayCount, ControlPoints)

```

```

SketchSegment ICreateSplineByEqnParams(
    ref System.int Properties,
    System.int KnotArrayCount,
    ref System.double Knots,
    System.int ControlPointArrayCount,
    ref System.double ControlPoints
)

```

```

SketchSegment^ ICreateSplineByEqnParams(
    & System.int% Properties,
    & System.int KnotArrayCount,
    & System.double% Knots,
    & System.int ControlPointArrayCount,
    & System.double% ControlPoints
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Properties

- in-process, unmanaged C++: Pointer to an array of properties including dimension, order, number of control points, periodicity (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

KnotArrayCount

Number of knots

Knots

- in-process, unmanaged C++: Pointer to an array of knots (e.g., knot1, knot2, and so on)
- VBA, VB.NET, C#, and C++/CLI: Not supported

ControlPointArrayCount

Number of control points

ControlPoints

- in-process, unmanaged C++: Pointer to an array of control points (e.g., controlpoint1[dimension], controlpoint2[dimension], and so on)
- VBA, VB.NET, C#, and C++/CLI: Not supported

Return Value

[Sketch segment](#) for the spline

Remarks

The Properties array contains four longs (placed in the first four doubles in the array):

- Dimension
- Order
- Number of Control Points
- Periodicity (true or false)

The Knots array contains doubles with (Number of Control Points + Order) elements.

The size of the ControlPoints array is based upon the curve dimension.

- Dimension = 2 then each Control Point is an array of 2 doubles (x, y)
- Dimension = 3 then each Control Point is an array of 3 doubles (x, y, z)
- Dimension = 4 then each Control Point is an array of 4 doubles (x, y, z, w) where w = weight

The parameters are provided as three arrays.

Pass the control point coordinates to this routine in sketch space. The Z value is interpreted as 0. In certain situations, you must transform your B-curve parameters to sketch space using [ISketch::ModelToSketchTransform](#).

NOTE: The curve can be either periodic or non-periodic. If the generated spline is a closed spline, then it must be flagged as periodic. Additionally, splines generated in sketches must be G1 continuous. If the data passed to this method does not generate a G1 continuous spline, then it is rejected and a spline is not created. If your data is not G1 continuous, then you must split the spline into multiple G1 segments and call this method for each segment.

This method does not work with [ISketchManager::AddToDB](#) or [ISketchManager::DisplayWhenAdded](#). It always adds the spline directly to the database (as if [ISketchManager::AddToDB\(True\)](#) is in effect), and you must redraw your document window to see the entities that you added (as if [ISketchManager::DisplayWhenAdded\(False\)](#) is in effect).

To create 3D splines, see [IModelDoc2::InsertCurveFilePoint](#) and related functions.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateSpline Method](#)
[ISketchManager::CreateSplineByEqnParams Method](#)
[ISketchManager::CreateSplinesByEqnParams Method](#)
[ISketchManager::ICreateSpline Method](#)
[ISketchManager::ICreateSplinesByEqnParams Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ICreateSplinesByEqnParams Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : ICreateSplinesByEqnParams Method (ISketchManager)

Creates one or more spline segments using the B-curve parameters provided.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateSplinesByEqnParams( _  
    ByRef Properties As System.Integer, _  
    ByVal KnotArrayCount As System.Integer, _  
    ByRef Knots As System.Double, _  
    ByVal ControlPointArrayCount As System.Integer, _  
    ByRef ControlPoints As System.Double _  
    ) As EnumSketchSegments
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim Properties As System.Integer  
Dim KnotArrayCount As System.Integer  
Dim Knots As System.Double  
Dim ControlPointArrayCount As System.Integer  
Dim ControlPoints As System.Double  
Dim value As EnumSketchSegments
```

```
value = instance.ICreateSplinesByEqnParams(Properties, KnotArrayCount, Knots,  
ControlPointArrayCount, ControlPoints)
```

C#

```
EnumSketchSegments ICreateSplinesByEqnParams(
    ref System.int Properties,
    System.int KnotArrayCount,
    ref System.double Knots,
    System.int ControlPointArrayCount,
    ref System.double ControlPoints
)
```

C++/CLI

```
EnumSketchSegments^ ICreateSplinesByEqnParams(
    & System.int% Properties,
    & System.int KnotArrayCount,
    & System.double% Knots,
    & System.int ControlPointArrayCount,
    & System.double% ControlPoints
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Properties

- in-process, unmanaged C++: Pointer to an array of dimension, order, number of control points, periodicity (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

KnotArrayCount

Number of knots

Knots

- in-process, unmanaged C++: Pointer to an array of knots (e.g., knot1, knot2, etc.)
- VBA, VB.NET, C#, and C++/CLI: Not supported

ControlPointArrayCount

Number of control points

ControlPoints

- in-process, unmanaged C++: Pointer to an array of control points (e.g., controlpoint1[dimension], controlpoint2[dimension], etc.)
- VBA, VB.NET, C#, and C++/CLI: Not supported

Return Value

- in-process, unmanaged C++: Pointer to an array of [sketch segments](#) of the newly created splines
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

The Properties array contains four longs (placed in the first four doubles in the array):

- Dimension
- Order
- Number of Control Points
- Periodicity (true or false)

The Knots array contains doubles with (Number of Control Points + Order) elements.

The size of the ControlPoints array is based upon the curve dimension.

- Dimension = 2 then each Control Point is an array of 2 doubles (x, y)
- Dimension = 3 then each Control Point is an array of 3 doubles (x, y, z)
- Dimension = 4 then each Control Point is an array of 4 doubles (x, y, z, w) where w = weight

The parameters are provided as three arrays.

Pass the control point coordinates to this routine in sketch space. The Z value is interpreted as 0. In certain situations, you must transform your B-curve parameters to sketch space using [ISketch::ModelToSketchTransform](#).

NOTE: The curve can be either periodic or non-periodic. If the generated spline is a closed spline, then it must be flagged as periodic. Additionally, splines generated in sketches must be G1 continuous. If the data passed to this method does not generate a G1 continuous spline, then it is rejected and a spline is not created. If your data is not G1 continuous, then you must split the spline into multiple G1 segments and call this method for each segment.

This method does not work with [ISketchManager::AddToDB](#) or [ISketchManager::DisplayWhenAdded](#). It always adds the spline directly to the database (as if [ISketchManager::AddToDB\(True\)](#) is in effect), and you must redraw your document window to see the entities that you added (as if [ISketchManager::DisplayWhenAdded\(False\)](#) is in effect).

To create 3D splines, see [IModelDoc2::InsertCurveFilePoint](#) and related functions.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateSpline Method](#)
[ISketchManager::CreateSplineByEqnParams Method](#)
[ISketchManager::CreateSplinesByEqnParams Method](#)
[ISketchManager::ICreateSpline Method](#)
[ISketchManager::ICreateSplineByEqnParams Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSketchBlockDefinitions Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : IGetSketchBlockDefinitions Method (ISketchManager)

Gets all of the block definitions.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSketchBlockDefinitions( _  
    ByVal BlockDefCount As System.Integer _  
) As SketchBlockDefinition
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim BlockDefCount As System.Integer  
Dim value As SketchBlockDefinition  
  
value = instance.IGetSketchBlockDefinitions(BlockDefCount)
```

C#

```
SketchBlockDefinition IGetSketchBlockDefinitions(  
    System.int BlockDefCount  
)
```

C++/CLI

```
SketchBlockDefinition^ IGetSketchBlockDefinitions(  
& System.int BlockDefCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BlockDefCount
Number of block definitions

Return Value

- in-process, unmanaged C++: Pointer to an array of [block definitions](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISketchManager::GetSketchBlockDefinitionCount](#) to get the value of BlockDefCount.

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::GetSketchBlockDefinitions Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InferenceMode Property (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : InferenceMode Property (ISketchManager)

Obsolete. Superseded by [ISldWorks::GetUserPreferenceToggle](#) or [ISldWorks::SetUserPreferenceToggle](#) and [swUserPreferenceToggle_e.swSketchInference](#).

.NET Syntax

Visual Basic (Declaration)

Property InferenceMode As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim value As System.Boolean
```

```
instance.InferenceMode = value
```

```
value = instance.InferenceMode
```

C#

```
System.bool InferenceMode {get; set;}
```

C++/CLI

```
property System.bool InferenceMode {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if sketch inference mode is on, false if off

Visual Basic for Applications (VBA) Syntax

See [SketchManager::InferenceMode](#).

Remarks

This property affects sketch entity snapping and inferring constraints to other geometry during creation.

Setting inference mode to...	Allows...
True	Inferencing during sketch operations, subject to other settings that may disable inferencing such as ISketchManager::AutoInference , ISketchManager::AddToDB , and IDrawingDoc::StartDrawing .
False	Faster sketching without inferencing, similar to SketchManager::AddToDB, except that using this method does not disable undo operations.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

Insert3DSketch Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : Insert3DSketch Method (ISketchManager)

Inserts a new 3D sketch in a model or closes the active sketch.

.NET Syntax

Visual Basic (Declaration)

```
Sub Insert3DSketch( _  
    ByVal UpdateEditRebuild As System.Boolean _  
)
```

Visual Basic (Usage)

```
Dim instance As ISketchManager  
Dim UpdateEditRebuild As System.Boolean  
  
instance.Insert3DSketch(UpdateEditRebuild)
```

C#

```
void Insert3DSketch(  
    System.bool UpdateEditRebuild  
)
```

C++/CLI

```
void Insert3DSketch(  
&    System.bool UpdateEditRebuild  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UpdateEditRebuild
True if you want to edit and rebuild, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::Insert3DSketch](#).

Example

[Insert Explode Line Sketch and Jog Line \(VB.NET\)](#)
[Insert Explode Line Sketch and Jog Line \(VBA\)](#)
[Insert Explode Line Sketch and Jog Line \(C#\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::AddToDB Property](#)
[ISketchManager::CreateSketchPlane Method](#)
[ISketchManager::DisplayWhenAdded Property](#)
[ISketchManager::InsertExplodeLineSketch Method](#)
[ISketchSegment::JogLine Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

InsertExplodeLineSketch Method (ISketchManager)

[See Also](#) [Example](#)



Inserts or closes an explode line sketch in an exploded view.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertExplodeLineSketch() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim value As System.Boolean

value = instance.InsertExplodeLineSketch()
```

C#

```
System.bool InsertExplodeLineSketch()
```

C++/CLI

```
System.bool InsertExplodeLineSketch();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the explode line sketch is inserted or closed, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager.InsertExplodeLineSketch](#).

Example

[Insert Explode Line Sketch and Route Line \(VB.NET\)](#)
[Insert Explode Line Sketch and Route Line \(VBA\)](#)
[Insert Explode Line Sketch and Jog Line \(VB.NET\)](#)
[Insert Explode Line Sketch and Jog Line \(VBA\)](#)
[Insert Explode Line Sketch and Jog Line \(C#\)](#)
[Insert Explode Line Sketch and Route Line \(C#\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[IAssemblyDoc::AutoExplode Method](#)
[ISketch::InsertRouteLine Method](#)
[ISketchManager::Insert3DSketch Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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InsertSketch Method (ISketchManager)

See Also [Example](#)



SolidWorks.Interop.sldworks Namespace > [ISketchManager Interface](#) : InsertSketch Method (ISketchManager)

Inserts a new sketch in the current part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Sub InsertSketch( _
    ByVal UpdateEditRebuild As System.Boolean _
)
```

Visual Basic (Usage)

```
Dim instance As ISketchManager  
Dim UpdateEditRebuild As System.Boolean  
  
instance.InsertSketch(UpdateEditRebuild)
```

C#

```
void InsertSketch(  
    System.bool UpdateEditRebuild  
)
```

C++/CLI

```
void InsertSketch(  
& System.bool UpdateEditRebuild  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UpdateEditRebuild

True to rebuild the part with any changes made to the sketch and exit sketch mode, false to not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::InsertSketch](#).

Example

[Insert Sheet Metal Edge Flange \(VBA\)](#)
[Create Equation-driven Curve \(C#\)](#)
[Create Equation-driven Curve \(VB.NET\)](#)
[Create Equation-driven Curve \(VBA\)](#)
[Insert and Resize Sketch Slot \(C#\)](#)
[Insert and Resize Sketch Slot \(VB.NET\)](#)
[Insert and Resize Sketch Slot \(VBA\)](#)

Remarks

This method does not support drawing documents.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSketchBlockInstance Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : InsertSketchBlockInstance Method (ISketchManager)

Inserts a block instance at the specified location using the block definition.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSketchBlockInstance( _
    ByVal BlockDef As SketchBlockDefinition, _
    ByVal Position As MathPoint, _
    ByVal Scale As System.Double, _
    ByVal Angle As System.Double _
) As SketchBlockInstance
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim BlockDef As SketchBlockDefinition
Dim Position As MathPoint
Dim Scale As System.Double
Dim Angle As System.Double
Dim value As SketchBlockInstance

value = instance.InsertSketchBlockInstance(BlockDef, Position, Scale, Angle)
```

C#

```
SketchBlockInstance InsertSketchBlockInstance(
    SketchBlockDefinition BlockDef,
    MathPoint Position,
    System.double Scale,
    System.double Angle
)
```

C++/CLI

```
SketchBlockInstance^ InsertSketchBlockInstance(
    & SketchBlockDefinition^ BlockDef,
    & MathPoint^ Position,
    & System.double Scale,
    & System.double Angle
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BlockDef
[Block](#) for this block instance
Position
[Position](#) of this block instance
Scale
 Scale for this block instance
Angle
 Rotation angle for this block instance

Return Value

[Block instance](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::InsertSketchBlockInstance](#).

Example

[Create Block Definition and Insert Block Instance \(VBA\)](#)
[Create Block Definition and Insert Block Instance \(C#\)](#)
[Create Block Definition and Insert Block Instance \(VB.NET\)](#)

Remarks

This method creates a block definition if the block definition does not exist.

- or -

If the definition exists, then this method uses that block definition to create the block instance. The name of the block instance is the same as the filename of the block file, without the filename extension.

If the entities of a block are associated with one or more layers and those layers do not already exist in the drawing, then the layers are inserted in the drawing and the associations between the entities of the block and the layers are maintained.

The block instance is inserted on the current drawing layer.

TIP: If inserting the same sketch block multiple times, do not create the block definition more than once. Instead, use the same block definition for each call to `SketchManager::InsertSketchBlockInstance`.

To save a block instance and its definition into a block file (`.sldblk`), use [ISketchBlockDefinition::Save](#).

This method does not work for drawings opened in view-only mode.

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSketchPicture Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : InsertSketchPicture Method (ISketchManager)

Obsolete. Superseded by [ISketchManager::InsertSketchPicture2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSketchPicture( _  
    ByVal FileName As System.String _  
) As SketchPicture
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim FileName As System.String  
Dim value As SketchPicture
```

```
value = instance.InsertSketchPicture(FileName)
```

C#

```
SketchPicture InsertSketchPicture(  
    System.string FileName  
)
```

C++/CLI

```
SketchPicture^ InsertSketchPicture(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Path to image file including file extension

Return Value

[Picture](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager.InsertSketchPicture](#).

See Also

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertSketchPicture2 Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : InsertSketchPicture2 Method (ISketchManager)

Inserts a picture on the current drawing sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSketchPicture2( _  
    ByVal FileName As System.String, _  
    ByVal HighestResolution As System.Boolean _  
) As SketchPicture
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim FileName As System.String  
Dim HighestResolution As System.Boolean  
Dim value As SketchPicture  
  
value = instance.InsertSketchPicture2(FileName, HighestResolution)
```

C#

```
SketchPicture InsertSketchPicture2(  
    System.string FileName,  
    System.bool HighestResolution  
)
```

C++/CLI

```
SketchPicture^ InsertSketchPicture2(  
& System.String^ FileName,  
& System.bool HighestResolution  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Path to image file including file extension

HighestResolution

True to insert images up to 8192 pixels without compression, false to compress images to 2048 pixels before insertion (see **Remarks**)

Return Value

[Picture](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::InsertSketchPicture2](#).

Example

[Flip Sketch Picture \(VBA\)](#)
[Flip Sketch Picture \(VB.NET\)](#)
[Flip Sketch Picture \(C#\)](#)

Remarks

If the document type is not a drawing, then HighestResolution defaults to false.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2019 SP04, Revision Number 27.4

SOLIDWORKS API Help

[Send comments](#) on this topic.

IntersectCurves Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : IntersectCurves Method (ISketchManager)

Creates a sketched intersection curve.

.NET Syntax

Visual Basic (Declaration)

```
Sub IntersectCurves()
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
instance.IntersectCurves()
```

C#

```
void IntersectCurves()
```

C++/CLI

```
void IntersectCurves();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SketchManager.IntersectCurves](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

MakeSketchBlockFromFile Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : MakeSketchBlockFromFile Method (ISketchManager)

Creates a block definition using the specified file.

.NET Syntax

Visual Basic (Declaration)

```
Function MakeSketchBlockFromFile( _  
    ByVal InsertionPoint As MathPoint, _
```

```

    ByVal FileName As System.String, _
    ByVal LinkedToFile As System.Boolean, _
    ByVal Scale As System.Double, _
    ByVal Angle As System.Double _
) As SketchBlockDefinition

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim InsertionPoint As MathPoint
Dim FileName As System.String
Dim LinkedToFile As System.Boolean
Dim Scale As System.Double
Dim Angle As System.Double
Dim value As SketchBlockDefinition

value = instance.MakeSketchBlockFromFile(InsertionPoint, FileName, LinkedToFile, Scale, Angle)

```

C#

```

SketchBlockDefinition MakeSketchBlockFromFile(
    MathPoint InsertionPoint,
    System.string FileName,
    System.bool LinkedToFile,
    System.double Scale,
    System.double Angle
)

```

C++/CLI

```

SketchBlockDefinition^ MakeSketchBlockFromFile(
    & MathPoint^ InsertionPoint,
    & System.String^ FileName,
    & System.bool LinkedToFile,
    & System.double Scale,
    & System.double Angle
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InsertionPoint
[Insertion point](#), which is a 2D point with z = 0.0, for the block definition
FileName
 Name of the external file to use to create the block definition
LinkedToFile
 True to link the block definition to the file, false to not
Scale
 Scale for the block definition
Angle
 Rotation angle for the block definition

Return Value

[Block definition](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::MakeSketchBlockFromFile](#).

Remarks

If the entities of a block are associated with one or more layers and those layers do not already exist in the drawing, then the layers are inserted in the drawing and the associations between the entities of the block and the layers are maintained.

The block instance is inserted on the current drawing layer.

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

[ISketchManager::MakeSketchBlockFromSelected Method](#)
[ISketchManager::MakeSketchBlockFromSketch Method](#)
[ISketchBlockDefinition::FileName Property](#)
[ISketchBlockDefinition::LinkToFile Property](#)
[ISketchBlockDefinition::Save Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MakeSketchBlockFromSelected Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : MakeSketchBlockFromSelected Method (ISketchManager)

Creates a block definition at the specified location from the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Function MakeSketchBlockFromSelected( _  
    ByVal InsertionPoint As MathPoint _  
) As SketchBlockDefinition
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim InsertionPoint As MathPoint  
Dim value As SketchBlockDefinition  
  
value = instance.MakeSketchBlockFromSelected(InsertionPoint)
```

C#

```
SketchBlockDefinition MakeSketchBlockFromSelected(  
    MathPoint InsertionPoint  
)
```

C++/CLI

```
SketchBlockDefinition^ MakeSketchBlockFromSelected(  
& MathPoint^ InsertionPoint  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InsertionPoint
[Insertion point](#), which is a 2D point with z = 0.0, for the block definition

Return Value

[Block definition](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::MakeSketchBlockFromSelected](#).

Example

[Create Block Definition and Insert Block Instance \(VBA\)](#)
[Create Block Definition and Insert Block Instance \(C#\)](#)
[Create Block Definition and Insert Block Instance \(VB.NET\)](#)

Remarks

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::MakeSketchBlockFromFile Method](#)
[ISketchManager::MakeSketchBlockFromSketch Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MakeSketchBlockFromSketch Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : MakeSketchBlockFromSketch Method (ISketchManager)

Creates a block definition at the specified location using all of the sketch entities in the active sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function MakeSketchBlockFromSketch( _  
    ByVal InsertionPoint As MathPoint, _  
    ByVal Sketch As Sketch _  
) As SketchBlockDefinition
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim InsertionPoint As MathPoint  
Dim Sketch As Sketch  
Dim value As SketchBlockDefinition  
  
value = instance.MakeSketchBlockFromSketch(InsertionPoint, Sketch)
```

C#

```
SketchBlockDefinition MakeSketchBlockFromSketch(  
    MathPoint InsertionPoint,  
    Sketch Sketch  
)
```

C++/CLI

```
SketchBlockDefinition^ MakeSketchBlockFromSketch(  
& MathPoint^ InsertionPoint,  
& Sketch^ Sketch  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[InsertionPoint](#)
[Insertion point](#), which is a 2D point with z = 0.0, for the block definition
[Sketch](#)
[Sketch](#) to use

Return Value

[Block definition](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::MakeSketchBlockFromSketch](#).

Remarks

See [Block Definitions and Block Instances](#) for details.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::MakeSketchBlockFromFile Method](#)
[ISketchManager::MakeSketchBlockFromSelected Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MakeSketchChain Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : MakeSketchChain Method (ISketchManager)

Creates a sketch path using the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Function MakeSketchChain() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim value As System.Boolean  
  
value = instance.MakeSketchChain()
```

C#

```
System.bool MakeSketchChain()
```

C++/CLI

```
System.bool MakeSketchChain();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the a sketch path is created, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::MakeSketchChain](#).

Example

[Create Sketch Path \(VBA\)](#)
[Create Sketch Path \(VB.NET\)](#)
[Create Sketch Path \(C#\)](#)

Remarks

This method is not valid:

- unless a sketch is in edit mode.
- for 3D sketches.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

PerimeterCircle Method (ISketchManager)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : PerimeterCircle Method (ISketchManager)

Draws a 3-point perimeter arc.

 .NET Syntax

Visual Basic (Declaration)

```
Function PerimeterCircle( _
    ByVal X1 As System.Double, _
    ByVal Y1 As System.Double, _
    ByVal X2 As System.Double, _
    ByVal Y2 As System.Double, _
    ByVal X3 As System.Double, _
    ByVal Y3 As System.Double _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim X1 As System.Double
Dim Y1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim X3 As System.Double
Dim Y3 As System.Double
Dim value As System.Object

value = instance.PerimeterCircle(X1, Y1, X2, Y2, X3, Y3)
```

C#

```
System.object PerimeterCircle(
    System.double X1,
    System.double Y1,
    System.double X2,
    System.double Y2,
    System.double X3,
    System.double Y3
)
```

C++/CLI

```
System.Object^ PerimeterCircle(
    & System.double X1,
    & System.double Y1,
    & System.double X2,
    & System.double Y2,
    & System.double X3,
    & System.double Y3
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X1](#)
coordinate for first point

[Y1](#)
y coordinate for first point

[X2](#)
x coordinate for second point

[Y2](#)
y coordinate for second point

[X3](#)
x coordinate for third point

Y3
y coordinate for third point

Return Value

[3-point_perimeter_arc](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::PerimeterCircle](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ReverseEndPointTangent Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : ReverseEndPointTangent Method (ISketchManager)

Reverses the end point tangent direction of splines and arcs.

.NET Syntax

Visual Basic (Declaration)

```
Function ReverseEndPointTangent( _  
    ByVal ForceDeleteConstraints As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim ForceDeleteConstraints As System.Boolean  
Dim value As System.Integer  
  
value = instance.ReverseEndPointTangent(ForceDeleteConstraints)
```

C#

```
System.int ReverseEndPointTangent(  
    System.bool ForceDeleteConstraints  
)
```

C++/CLI

```
System.int ReverseEndPointTangent(  
    & System.bool ForceDeleteConstraints  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ForceDeleteConstraints
True to force the deletion of constraints, false to not

Return Value

Result code as defined in [swReverseEndPointTangentResult_e](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::ReverseEndPointTangent](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RotateOrCopy3DAboutVector Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : RotateOrCopy3DAboutVector Method (ISketchManager)

Rotates, and optionally copies, the selected 3D sketch entities about the specified vector.

.NET Syntax

Visual Basic (Declaration)

```
Function RotateOrCopy3DAboutVector( _  
    ByVal Copy As System.Boolean, _  
    ByVal NumCopies As System.Integer, _  
    ByVal KeepRelations As System.Boolean, _  
    ByVal BaseX As System.Double, _  
    ByVal BaseY As System.Double, _  
    ByVal BaseZ As System.Double, _  
    ByVal VectorXCoord As System.Double, _  
    ByVal VectorYCoord As System.Double, _  
    ByVal VectorZCoord As System.Double, _  
    ByVal AngleAboutVector As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim Copy As System.Boolean  
Dim NumCopies As System.Integer  
Dim KeepRelations As System.Boolean  
Dim BaseX As System.Double  
Dim BaseY As System.Double  
Dim BaseZ As System.Double  
Dim VectorXCoord As System.Double  
Dim VectorYCoord As System.Double  
Dim VectorZCoord As System.Double  
Dim AngleAboutVector As System.Double  
Dim value As System.Boolean
```

```
value = instance.RotateOrCopy3DAboutVector(Copy, NumCopies, KeepRelations, BaseX, BaseY, BaseZ,  
VectorXCoord, VectorYCoord, VectorZCoord, AngleAboutVector)
```

C#

```
System.bool RotateOrCopy3DAboutVector(  
    System.bool Copy,  
    System.int NumCopies,  
    System.bool KeepRelations,  
    System.double BaseX,  
    System.double BaseY,  
    System.double BaseZ,  
    System.double VectorXCoord,  
    System.double VectorYCoord,  
    System.double VectorZCoord,  
    System.double AngleAboutVector  
)
```

C++/CLI

```

System.bool RotateOrCopy3DAboutVector(
& System.bool Copy,
& System.int NumCopies,
& System.bool KeepRelations,
& System.double BaseX,
& System.double BaseY,
& System.double BaseZ,
& System.double VectorXCoord,
& System.double VectorYCoord,
& System.double VectorZCoord,
& System.double AngleAboutVector
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Copy
True to copy the 3D sketch entities, false to not

NumCopies
Number of copies

KeepRelations
True to keep sketch relations, false to not

BaseX
X coordinate of the base point from which to rotate the 3D sketch entities

BaseY
Y coordinate of the base point from which to rotate the 3D sketch entities

BaseZ
Z coordinate of the base point from which to rotate the 3D sketch entities

VectorXCoord
X vector component of the axis of rotation, parallel to the base point

VectorYCoord
Y vector component of the axis of rotation, parallel to the base point

VectorZCoord
Z vector component of the axis of rotation, parallel to the base point

AngleAboutVector
Angle at which to rotate the sketch entities

Return Value

True if the 3D sketch entities rotated and are optionally copied about the specified vector, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::RotateOrCopy3DAboutVector](#).

Example

[Rotate and Copy 3D Sketch About Vector \(VB.NET\)](#)
[Rotate and Copy 3D Sketch About Vector \(VBA\)](#)
[Rotate and Copy 3D Sketch About Vector \(C#\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[IModelDocExtension::RotateOrCopy Method](#)
[ISketchManager::RotateOrCopy3DAboutXYZ Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RotateOrCopy3DAboutXYZ Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : RotateOrCopy3DAboutXYZ Method (ISketchManager)

Rotates, and optionally copies, the selected 3D sketch entities about the specified x, y, and z coordinates.

.NET Syntax

Visual Basic (Declaration)

```

Function RotateOrCopy3DAboutXYZ( _
    ByVal Copy As System.Boolean, _

```

```

ByVal NumCopies As System.Integer, _
ByVal KeepRelations As System.Boolean, _
ByVal BaseX As System.Double, _
ByVal BaseY As System.Double, _
ByVal BaseZ As System.Double, _
ByVal AngleAboutXDir As System.Double, _
ByVal AngleAboutYDir As System.Double, _
ByVal AngleAboutZDir As System.Double _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim Copy As System.Boolean
Dim NumCopies As System.Integer
Dim KeepRelations As System.Boolean
Dim BaseX As System.Double
Dim BaseY As System.Double
Dim BaseZ As System.Double
Dim AngleAboutXDir As System.Double
Dim AngleAboutYDir As System.Double
Dim AngleAboutZDir As System.Double
Dim value As System.Boolean

```

```

value = instance.RotateOrCopy3DAboutXYZ(Copy, NumCopies, KeepRelations, BaseX, BaseY, BaseZ,
AngleAboutXDir, AngleAboutYDir, AngleAboutZDir)

```

C#

```

System.bool RotateOrCopy3DAboutXYZ(
    System.bool Copy,
    System.int NumCopies,
    System.bool KeepRelations,
    System.double BaseX,
    System.double BaseY,
    System.double BaseZ,
    System.double AngleAboutXDir,
    System.double AngleAboutYDir,
    System.double AngleAboutZDir
)

```

C++/CLI

```

System.bool RotateOrCopy3DAboutXYZ(
    & System.bool Copy,
    & System.int NumCopies,
    & System.bool KeepRelations,
    & System.double BaseX,
    & System.double BaseY,
    & System.double BaseZ,
    & System.double AngleAboutXDir,
    & System.double AngleAboutYDir,
    & System.double AngleAboutZDir
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Copy

True to copy 3D sketch entities, false to not

NumCopies

Number of copies

KeepRelations

True to keep sketch relations, false to not

BaseX

X coordinate from the base point from which to rotate the 3D sketch entities

BaseY

Y coordinate from the base point from which to rotate the 3D sketch entities

BaseZ

Z coordinate from the base point from which to rotate the 3D sketch entities

AngleAboutXDir

Angle to rotate the 3D sketch entities in the x direction

AngleAboutYDir

Angle to rotate 3D sketch entities in the y direction

AngleAboutZDir

Angle to rotate 3D sketch entities in the z direction

Return Value

True if the 3D sketch entities are rotated, and optionally copied, in the specified x, y, and z directions, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::RotateOrCopy3DAboutXYZ](#).

Example

[Rotate and Copy 3D Sketch About Coordinates \(C#\)](#)
[Rotate and Copy 3D Sketch about Coordinates \(VB.NET\)](#)
[Rotate and Copy 3D Sketch About Coordinates \(VBA\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::RotateOrCopy3DAboutVector Method](#)
[IModelDocExtension::RotateOrCopy Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0